The first version of ASP.NET was released in 2002 as part of .NET Framework 1.0. The ASP.NET Web Forms paradigm that it introduced differed significantly from the conventional scripting environments of classic ASP and PHP. ASP.NET Web Forms allowed developers to create web applications rapidly by using a graphical designer and a simple event model that mirrored desktop application-building techniques.

The ASP.NET framework allowed developers to create new applications quickly, but over time the web development ecosystem changed. It became apparent that ASP.NET Web Forms suffered from many problems, especially in building larger applications. In particular, a lack of testability, a complex stateful model, and limited influence on the generated HTML (making client-side development difficult) led developers to evaluate other options.

In response, Microsoft released the first version of ASP.NET MVC in 2009, based on the Model-View-Controller (MVC) pattern, a common web pattern used in frameworks such as Ruby on Rails, Django, and Java Spring. This framework allowed developers to separate UI elements from application logic, made testing easier, and provided tighter control of the HTML-generation process.

ASP.NET MVC has been through four more iterations since its first release, but all these iterations were built on the same underlying framework provided by the System .Web.dll file. This library is part of .NET Framework, so it comes preinstalled with all versions of Windows. It contains all the core code that ASP.NET uses when you build a web application.

This dependency brings both advantages and disadvantages. On one hand, the ASP.NET framework is a reliable, battle-tested platform that’s fine for building web applications in Windows. It provides a wide range of features that have been in production for many years, and it’s well known by virtually all Windows web developers.

On the other hand, this reliance is limiting. Changes to the underlying System.Web.dll file are far-reaching and, consequently, slow to roll out, which limits the extent to which ASP.NET is free to evolve and results in release cycles happening only every few years. There’s also an explicit coupling with the Windows web host, Internet Information Services (IIS), which precludes its use on non-Windows platforms.

More recently, Microsoft declared .NET Framework to be “done.” It won’t be removed or replaced, but it also won’t receive any new features. Consequently, ASP.NET based on System.Web.dll won’t receive new features or updates either.

In recent years, many web developers have started looking at cross-platform web frameworks that can run on Windows as well as Linux and macOS. Microsoft felt the time had come to create a framework that was no longer tied to its Windows legacy; thus, ASP.NET Core was born.